Majorana sums of squares and rectangular matrix multiplication: noncommutative circuits for the identity polynomial of size $N^{1.6692}$

Carson Weitnauer

October 29, 2026

Abstract

We prove that Hrubeš's identity polynomial $\mathrm{ID}_N = \sum_{i,j\in[N]} x_i y_j x_i y_j$ in $2N$ noncommuting variables, the noncommutative elementary symmetric polynomial $S_{4,N}$, and the rectangular permanent $\mathrm{perm}_{4,N}$ have noncommutative arithmetic circuits of size $O(N^{1.6692+\varepsilon})$ for every $\varepsilon > 0$, over every field of characteristic different from two. The previous published bound is $O(N^{1.96})$, due to Hrubeš. Without fast matrix multiplication, a different construction gives $N^{1.72790+o(1)}$, also over every field of characteristic different from two.

The construction combines three ingredients: a weighted sum-of-squares identity over the bilinear forms $X^{\mathsf{T}} P_\nu Y$ of real Pauli words P_ν, whose weights are supported on a "corner" of the Jordan–Wigner (Majorana) classes; blocks of Majorana modes that are moved by Clifford conjugation so that all forms of a block are recovered by Walsh transforms of one rectangular matrix product; and a rectangular matrix-multiplication bound. The exponent is $2\mathrm{H}(\frac{1}{4}) - 2\mu\mathrm{H}(\frac{1}{4\mu}) + \mu w$, where w is an admissible exponent for multiplying $n \times n^{(1-\mu)/\mu}$ by $n^{(1-\mu)/\mu} \times n$ matrices. With Uznański's 2026 bound $\omega(1,k,1) \le 1 + k + \frac{1}{4k}$, in the integer form formalized by Zhou and Wang, this gives 1.66917. With OpenAI's 2026 bounds instead, an independent route gives $N^{1.690568+o(1)}$ in characteristic zero and in all but finitely many characteristics.

The theorems and lemmas stated here are formalized and machine-checked in Lean 4 with Mathlib, using Zhou and Wang's and OpenAI's Lean formalizations of the matrix-multiplication inputs and only Lean's standard axioms; the remarks and numerical comparisons are not.

1 Introduction

1.1 Background

Hurwitz's problem [12, 26, 28] asks for the smallest $s = \sigma(n)$ such that there is a sum-of-squares identity

$$(x_1^2 + \cdots + x_n^2)(y_1^2 + \cdots + y_n^2) = f_1^2 + \cdots + f_s^2 \tag{1}$$

with bilinear forms f_k. Hrubeš, Wigderson and Yehudayoff [11] connected this classical question to noncommutative arithmetic circuits, the model of Nisan [22]: a lower bound $\sigma(n) \ge n^{1+\varepsilon}$ would give an exponential lower bound for the noncommutative permanent. Their reduction runs through the identity polynomial

$$\mathrm{ID}_N = \sum_{i,j\in[N]} x_i y_j x_i y_j,$$

whose noncommutative circuit complexity is, up to constant factors, governed by sum-of-squares identities whose bilinear forms are cheap to compute together (Hrubeš [10, Lemma 22]). In 2024 Hrubeš proved the first truly subquadratic bound $\sigma(n) = O(n^{1.62})$ over $\mathbb{C}$ and, combining it with rectangular matrix multiplication, that ID_N, $S_{4,N}$ and $\mathrm{perm}_{4,N}$ have noncommutative circuits of size $O(N^{1.96})$ over every field of characteristic different from two [9, 10]. (His proof gives $O(N^{1.954})$ from the rectangular bound of Le Gall and Urrutia [17]. Run with n^r copies of the y variables, his argument gives the exponent $(r + \omega(1, r, 1) + \delta)/(r + 1)$ with $\delta < 0.616$; with Uznański's $\omega(1, 2, 1) \le \frac{25}{8}$ [33], the input behind Theorem 1.1, this is below 1.9137 at $r = 2$. This comparison is computed here, not formalized.) The bound $\sigma(n) \le n^{\sqrt{2}+o(1)}$ over $\mathbb{C}$ follows from Appendix B of Anthropic's October 2026 paper [3]; it was observed independently on October 27, 2026 by Kapila [14], by the author (with a Lean proof built on Anthropic's formalization [4], in the `theorem-b` directory of this paper's repository), and by Tulsyan [37]. Fed into Hrubeš's argument it improves the circuit exponent, but only to about 1.85–1.89 depending on the matrix-multiplication input. Before Hrubeš, the best bound for ID_N was the trivial $O(N^2)$, and branching programs of size about N^2 for $S_{4,N}$ and $\mathrm{perm}_{4,N}$ were given by Arvind, Chatterjee, Datta and Mukhopadhyay [2].

1.2 Results

Throughout, $\mathrm{size}^{\mathrm{nc}}_{\mathbb{F}}(f)$ is the size of a smallest noncommutative circuit over the field $\mathbb{F}$ computing f (Definition 2.1), and we write $\mathrm{size}^{\mathrm{nc}}_{\mathbb{F}}(f_N) \le N^{c+o(1)}$ for "$\mathrm{size}^{\mathrm{nc}}_{\mathbb{F}}(f_N) = O_\varepsilon(N^{c+\varepsilon})$ for every $\varepsilon > 0$".

Theorem 1.1 (main theorem)**.** *Over every field $\mathbb{F}$ of characteristic different from two,* $\mathrm{size}^{\mathrm{nc}}_{\mathbb{F}}(\mathrm{ID}_N)$*,* $\mathrm{size}^{\mathrm{nc}}_{\mathbb{F}}(S_{4,N})$ *and* $\mathrm{size}^{\mathrm{nc}}_{\mathbb{F}}(\mathrm{perm}_{4,N})$ *are at most* $N^{1.6692+o(1)}$*.*

The matrix-multiplication input is Uznański's rectangular bound [33], through the integer rank decompositions of Zhou and Wang [34]: one integer decomposition maps into every field, so no characteristic other than two is excluded. An independent route uses OpenAI's bounds instead.

Theorem 1.2 (OpenAI's inputs)**.** *Let* $c_0 = 211321/125000 = 1.690568$*.*

(a) Over every field $\mathbb{F}$ of characteristic zero, the three polynomials have circuits of size $N^{c_0+o(1)}$.

(b) There is a finite set S of primes such that the same holds over every field $\mathbb{F}$ with $\mathrm{char}\,\mathbb{F} \notin S \cup \{2\}$*.*

The set S in (b) comes from the Lefschetz principle and is not computed. Theorem 1.2 is weaker than Theorem 1.1, but its matrix-multiplication input is independent of it, so either suffices for a subquadratic bound below Theorem 1.3.

Theorem 1.3 (no fast matrix multiplication)**.** *Over every field of characteristic different from two,* $\mathrm{size}^{\mathrm{nc}}(\mathrm{ID}_N)$*,* $\mathrm{size}^{\mathrm{nc}}(S_{4,N})$ *and* $\mathrm{size}^{\mathrm{nc}}(\mathrm{perm}_{4,N})$ *are at most* $N^{1.727899+o(1)}$*.*

(The construction behind Theorem 1.3 has exact exponent $c_* = 1.72789810916\ldots$; the formal statement uses the rational upper bound 1.727899.) Its simplest precursor, a per-bit decomposition giving $N^{\log_2(12)/2} = N^{1.7925}$, realizes the single-bit identity for the crossing tensor and the "log 12 of 4 bits" count in Appendix H of [3]; Pauli (Bell) bases for the same tensor were also used by Tulsyan [37], for a different cost measure.

Theorem 1.3 is the author's earlier "Pauli cube" construction (October 28, 2026); we state it for comparison and refer to its formalization (Section 8). This paper proves Theorems 1.1 and 1.2. Their heart is the following conditional bound, which turns block oracles for rectangular products (Definition 4.2) into a circuit exponent.

Theorem 1.4 (the construction)**.** *Let $\mathbb{F}$ be a field with $2 \neq 0$ that contains a square root of -1. Let $\frac{1}{2} < \mu < 1$ and $w \geq 2$, and put $m = \lceil \mu n \rceil$. Suppose that for every $\varepsilon > 0$ there is C such that for every n there is a block oracle over $\mathbb{F}$ (Definition 4.2) for $2^m \times 2^{n-m}$ by $2^{n-m} \times 2^m$ products with cost at most $C(2^m)^{w+\varepsilon}$. Then*

$$\operatorname{size}_{\mathbb{F}}^{\mathrm{nc}}(\mathrm{ID}_N) \leq N^{C(\mu,w)+o(1)}, \qquad C(\mu, w) = 2\mathrm{H}\big(\tfrac{1}{4}\big) - 2\mu\,\mathrm{H}\Big(\frac{1}{4\mu}\Big) + \mu w,$$

where $\mathrm{H}(p) = -p\log_2 p - (1-p)\log_2(1-p)$.

A block oracle is more than a matrix-multiplication algorithm over $\mathbb{F}$: its inputs are noncommuting, and it must also evaluate a trilinear middle-insertion form. Two sources supply oracles of the required cost (Section 4): straight-line programs over $\mathbb{C}$, so that over $\mathbb{C}$ any admissible exponent w at aspect $\frac{1-\mu}{\mu}$ in the sense of Section 2.4 suffices; and a single rank decomposition over $\mathbb{F}$ (Lemma 7.1), the source Theorem 1.1 uses.

With Uznański's bound at $\mu = \frac{18}{29}$ (aspect $\frac{11}{18}$, $w = \frac{200}{99}$) one gets $C = 1.66916842\ldots$ (Section 7.1). With OpenAI's bounds $\omega(1, 0.465, 1) = 2$ and $\omega(1, 0.709, 1) < 2.092$ [24], interpolated at $\mu = 0.6337$, one gets $C < 1.6905678 < c_0$ (Section 7.2); OpenAI's thin endpoint alone ($\mu = 200/293$, $w = 2$) gives $C = 1.6938884\ldots$.

Remark 1.5. If $\omega(1, k, 1) = 2$ held for every $k \leq 1$, Theorem 1.4 would give exponents tending to $2\mathrm{H}(\frac{1}{4}) = 1.62255\ldots$ as $\mu \to \frac{1}{2}$, the growth exponent of the number of Pauli words in corner classes (Section 3). Since $\mathrm{H} \leq 1$, $C(\mu, w) \geq 2\mathrm{H}(\frac{1}{4})$ whenever $w \geq 2$, so this is the floor of the present cost formula. It is not a lower bound on circuit size.

1.3 Techniques

Corner sums of squares. A weighted identity $\sum_\nu w_\nu f_\nu^2 = (\sum X_x^2)(\sum Y_y^2)$ over the 4^n real Pauli bilinear forms $f_\nu(X, Y) = X^{\mathsf{T}} P_\nu Y$ holds if and only if a Fourier condition holds on the isotropic labels (Lemma 3.1). In Jordan–Wigner coordinates [13] a Pauli word is a product of Majorana operators, described by a symmetric part s and a skew part a. For weights depending only on the class $(|s|, |a|)$ the Fourier condition becomes a small linear system with Krawtchouk coefficients [15, 20]. We show it has a solution supported on the corner $\min(a, n-a) + \min(b, n-b) \leq n/2$ (Theorem 3.2); the proof is a duality argument in which a dual certificate would give a polynomial of degree at most $2n$ divisible by $(1 - X^4)^{\lfloor n/2 \rfloor + 1}$.

Majorana blocks. Swapping two Majorana modes is a conjugation by a single Clifford gate $1 + cP_u$. A block of $2m$ Majoranas can be moved onto the first m qubits, after which every word supported in the block (up to a fixed shift) has the form $P_{\alpha,\beta} \otimes I$. The corresponding bilinear forms are, after one shared change of basis of the inputs, Walsh coefficients along shifted diagonals of a single $2^m \times 2^m$ matrix product of $2^m \times 2^{n-m}$ and $2^{n-m} \times 2^m$ matrices. Hrubeš's middle-insertion term $f \star f^{yx}$ also collapses to a single trilinear form. So one block costs one rectangular matrix product plus $O(m4^m)$.

Covering and entropy. Each corner class is covered by blocks greedily, and an entropy calculation (Lemma 5.2) bounds the number of blocks by $2^{n(2\mathrm{H}(1/4) - 2\mu\mathrm{H}(1/(4\mu))) + o(n)}$.

Matrix multiplication and characteristic. Both routes reduce to one fixed rank decomposition of a matrix tensor $\langle a, b, a\rangle$, from which a recursive bilinear algorithm supplies every block product. Zhou and Wang's formalization provides integer decompositions along Uznański's curve, which work in every field. OpenAI's bounds are statements about straight-line programs over $\mathbb{C}$: a program of cost s gives a rank bound $2s$, ranks multiply under tensor products, and one fixed product decomposition is an existential first-order statement with integer coefficients, so it transfers by the Lefschetz principle.

1.4 Formal verification

The theorems above are formalized in Lean 4 [16] with Mathlib [21]. The formalization imports OpenAI's Lean library for its matrix-multiplication bounds [25] unchanged, and Zhou and Wang's Lean library [34] with two declaration-level changes needed by our stricter compiler settings, so the matrix-multiplication inputs are themselves machine-checked. The top-level Lean theorems, the definitions they use, and the checks performed are listed in Section 8.

2 Preliminaries

2.1 Noncommutative circuits

Definition 2.1 (Hrubeš [10])**.** A *noncommutative circuit* over $\mathbb{F}$ is a directed acyclic graph whose in-degree-zero nodes are labelled by variables or elements of $\mathbb{F}$, and whose other nodes have in-degree two and are labelled $+$ or $\times$, the two edges into a $\times$ node being labelled left and right. Each node computes a polynomial in the free algebra $\mathbb{F}\langle x_1, \dots\rangle$; the circuit computes f if some node does; its size is its number of nodes.

The polynomials of interest are ID_N above,

$$S_{4,N} = \sum_{(i_1,\dots,i_4)} x_{i_1}x_{i_2}x_{i_3}x_{i_4}, \qquad \mathrm{perm}_{4,N} = \sum_{(i_1,\dots,i_4)} x_{1,i_1}x_{2,i_2}x_{3,i_3}x_{4,i_4},$$

with sums over ordered quadruples of pairwise distinct elements of $[N]$. Hrubeš [10, Cor. 27] reduces $S_{4,N}$ and $\mathrm{perm}_{4,N}$ to ID_N with linear overhead (through ID^* and Proposition 26 there); we use these reductions as they are, so it suffices to bound $\mathrm{size}^{\mathrm{nc}}(\mathrm{ID}_N)$. Since setting variables to zero does not increase size, it also suffices to treat $N = 2^n$, indexing the variables by bit vectors $x \in \mathbb{F}_2^n$.

2.2 From sums of squares to circuits

For a bilinear form $f = \sum_{x,y} a_{xy}X_xY_y$ and a new variable z, Hrubeš [10] defines the noncommutative polynomials

$$f^{xy} = \sum a_{xy}X_xY_y, \quad f^{yx} = \sum a_{xy}Y_yX_x, \quad f \star z = \sum a_{xy}X_xzY_y, \quad f^{[2]} = \tfrac{1}{2}\big(f^{xy}f^{xy} + f \star f^{yx}\big).$$

Lemma 2.2 (Hrubeš, proof of [10, Lemma 22])**.** *Let $2 \neq 0$ in $\mathbb{F}$. If $(\sum_x X_x^2)(\sum_y Y_y^2) = \sum_k w_k f_k^2$ as commutative polynomials, with bilinear f_k, then $\sum_k w_k f_k^{[2]} = \mathrm{ID}$ in the free algebra.*

The cost of $\sum_k w_k f_k^{[2]}$ is the cost of the forms f_k^{xy} and f_k^{yx} plus that of the single trilinear form $\sum_k w_k f_k \star z_k$ evaluated at $z_k = f_k^{yx}$; Hrubeš bounds the latter by a Baur–Strassen argument [5]. Our blocks make all three cheap at once (Section 4).

2.3 Pauli forms in bit coordinates

A *word* is $\nu \in \{I, X, Z, J\}^n$, where $J = ZX = -XZ$; write $x_\nu, z_\nu \in \mathbb{F}_2^n$ for its X- and Z-parts. Its real Pauli matrix, indexed by $\mathbb{F}_2^n$, is

$$P_\nu[x, y] = [\, y = x + x_\nu \,] \, (-1)^{z_\nu \cdot x},$$

and we write $P_{\alpha,\beta}$ for the matrix with X-part α and Z-part β. These satisfy $P_{\alpha,\beta} P_{\alpha',\beta'} = (-1)^{\beta' \cdot \alpha} P_{\alpha+\alpha',\beta+\beta'}$. The *Pauli form* of ν is $f_\nu(X, Y) = X^{\mathsf{T}} P_\nu Y$.

2.4 Rectangular matrix multiplication

For $0 \le k \le 1$, an exponent τ is *admissible at aspect* k if for every $\varepsilon > 0$ there is C such that for every $n \ge 1$ some straight-line program (additions, subtractions, multiplications, constants) multiplies $n \times \lceil n^k \rceil$ by $\lceil n^k \rceil \times n$ matrices with at most $C n^{\tau+\varepsilon}$ operations. We write $\omega(1, k, 1)$ for the infimum of admissible exponents and $\alpha = \sup\{k : \omega(1, k, 1) = 2\}$. These are OpenAI's definitions [25], which our Lean development imports.

3 The corner sum of squares

Lemma 3.1 (Fourier criterion)**.** *Let $2 \neq 0$ in $\mathbb{F}$ and $w : \{\text{words}\} \to \mathbb{F}$. Put*

$$\hat{w}(t, b) = \sum_\nu w_\nu (-1)^{z_\nu \cdot t + x_\nu \cdot b}.$$

If $\hat{w}(t, b) = 2^n [t = b = 0]$ for all $t, b \in \mathbb{F}_2^n$ with $t \cdot b = 0$, then $(\sum_x X_x^2)(\sum_y Y_y^2) = \sum_\nu w_\nu f_\nu(X, Y)^2$.

Proof. Comparing coefficients of $X_x Y_y X_u Y_v$, the identity is equivalent to $\sum_\nu w_\nu (P_\nu[x, y] P_\nu[u, v] + P_\nu[x, v] P_\nu[u, y]) = 2[x = u,\ y = v]$. Both products vanish unless $d := x + u = y + v$, and then the left side is $g(x + y, d) + g(x + y + d, d)$ with $g(\alpha, d) = \sum_{x_\nu = \alpha} w_\nu (-1)^{z_\nu \cdot d}$. Taking the Fourier transform in α, the required identity $g(\alpha, d) + g(\alpha + d, d) = 2[d = 0]$ reads $\hat{w}(d, b)(1 + (-1)^{b \cdot d}) = 2 \cdot 2^n [d = 0][b = 0]$, which is automatic when $b \cdot d = 1$ and is the hypothesis when $b \cdot d = 0$. □

3.1 Majorana coordinates

Jordan–Wigner coordinates [13] write each word as $\nu = \nu(s, a)$ with $s, a \in \mathbb{F}_2^n$ via

$$x_\nu = s + a, \qquad (z_\nu)_j = a_j + \sum_{k > j} (s + a)_k.$$

This is a bijection; $P_{\nu(s,a)}$ is, up to sign, the product of the symmetric Majoranas $Z \cdots Z X_j$ over the support of s and the skew Majoranas $Z \cdots Z J_j$ over the support of a. The *class* of ν is $(|s|, |a|)$. In these coordinates the commutation form is $\langle (s, a), (s', a') \rangle = (|s| + |a|)(|s'| + |a'|) + s \cdot s' + a \cdot a'$, and the label (t, b) with Majorana coordinates (s', a') is isotropic ($t \cdot b = 0$) if and only if $q(|s'|, |a'|) := |a'| + \binom{|s'|+|a'|}{2} \equiv 0 \pmod 2$.

For class-constant weights $w_\nu = 2^n W(|s|, |a|)$, summing characters over a class gives Krawtchouk numbers $K_a(c) = [z^a](1 - z)^c (1 + z)^{n-c}$, and the Fourier criterion becomes the finite linear system

$$\sum_{a,b=0}^{n} W(a, b) \, (-1)^{(a+b)(c+d)} K_a(c) K_b(d) = [c = d = 0] \qquad (0 \le c, d \le n,\ q(c, d) = 0). \tag{2}$$

Theorem 3.2 (corner feasibility)**.** *Let $\mathbb{F}$ have $2 \neq 0$ and contain i with $i^2 = -1$. For every n, the system* (2) *has a solution W supported on the corner*

$$\mathcal{C}_n = \{(a, b) : \ \min(a, n-a) + \min(b, n-b) \leq \lfloor n/2 \rfloor\}.$$

Consequently there are weights on the 4^n words, supported on corner classes, with $(\sum X_x^2)(\sum Y_y^2) = \sum_\nu w_\nu f_\nu(X, Y)^2$.

Proof. If not, linear duality gives $y(c, d)$, supported on the pairs with $q(c, d) = 0$, with $y(0, 0) \neq 0$ and $\sum_{c,d} y(c, d)(-1)^{(a+b)(c+d)} K_a(c) K_b(d) = 0$ for every $(a, b) \in \mathcal{C}_n$. Put $P_a(X) = (1-X)^a(1+X)^{n-a} + (1+X)^a(1-X)^{n-a}$ and $\Psi = \sum_{a,b} \mathrm{coef}(a, b)\, P_a P_b$, where $\mathrm{coef}(a, b)$ is the left side above.

Ψ is divisible by $(1-X^2)^m$, $m = \lfloor n/2 \rfloor + 1$. For $(a, b) \in \mathcal{C}_n$ the coefficient vanishes; otherwise $((1-X)(1+X))^{\min(a,n-a)+\min(b,n-b)}$ divides $P_a P_b$, and the exponent is at least m.

Ψ is a multiple of $G = \sum_{c,d \text{ even}} y(c, d) X^{c+d}$. Exchanging the sums gives $\Psi = \sum_{c,d} y(c, d)\, S_{c,c+d} S_{d,c+d}$, where $S_{c,e} = \sum_a (-1)^{ae} K_a(c) P_a$. The homogenization identity $\sum_a K_a(c) s^a t^{n-a} = (t-s)^c (t+s)^{n-c}$ gives

$$S_{c,e} = \begin{cases} \big((2X)^c + (-2X)^c\big)\, 2^{n-c} & e \text{ even}, \\ 2^c \big((2X)^{n-c} + (-2X)^{n-c}\big) & e \text{ odd}, \end{cases}$$

so $S_{c,c+d} S_{d,c+d} = 4 \cdot 2^{2n} X^{c+d}$ when c and d are both even and 0 otherwise. Hence $\Psi = 4 \cdot 2^{2n} G$.

G is divisible by $(1+X^2)^m$. For even c, d, $q(c, d) = 0$ forces $4 \mid c + d$, so $G(iX) = G(X)$; applying $X \mapsto iX$ to $(1-X^2)^m \mid G$ gives $(1+X^2)^m \mid G$.

Since $2 \neq 0$, $(1-X^2)^m$ and $(1+X^2)^m$ are coprime, so $(1-X^4)^m \mid G$. But $\deg G \leq 2n < 4m$, so $G = 0$ and $y(0, 0) = G(0) = 0$, a contradiction. The second sentence follows from Lemma 3.1. □

The number of words in corner classes is $\sum_{(a,b) \in \mathcal{C}_n} \binom{n}{a} \binom{n}{b} = 2^{2\mathrm{H}(1/4)n + O(\log n)}$: the maximum is at $a, b \approx n/4$.

4 Majorana blocks

4.1 Moving Majoranas by conjugation

Index the $2n$ Majoranas by $k \in \{1, \ldots, 2n\}$ and identify a word with its Majorana vector $A \in \mathbb{F}_2^{2n}$ (the pair (s, a)); write P_A for its matrix. For $k \neq l$ let $u = e_k + e_l$, choose $c \in \{1, i\}$ with $(cP_u)^2 = -1$, and put $G_{kl} = 1 + cP_u$, so $G_{kl}^{-1} = \frac{1}{2}(1 - cP_u)$.

Lemma 4.1. *$G_{kl} P_A G_{kl}^{-1} = \lambda\, P_{\mathrm{swap}_{kl}(A)}$ for some $\lambda \neq 0$.*

Proof. P_u and P_A commute or anticommute according to $\langle u, A \rangle = A_k + A_l$. If they commute, conjugation fixes P_A and swap_{kl} fixes A. If they anticommute, $G P_A G^{-1} = -P_A\, cP_u = \lambda P_{A+u}$, and $A + u = \mathrm{swap}_{kl}(A)$. □

A *block* is a set U of $2m$ Majoranas; the *canonical block* T consists of the $2m$ Majoranas of the first m qubits. Composing at most $|U \setminus T| \leq 2m$ swaps gives a matrix Γ with $\Gamma P_A \Gamma^{-1} = \lambda_A P_{A'}$, A' supported in T, for every A supported in U. For a fixed *shift* $\sigma \in \mathbb{F}_2^{2n}$, $P_{A+\sigma} = \pm P_A P_\sigma$, hence

$$P_{A+\sigma} = \lambda\, \Gamma^{-1} P_{A'}\, (\Gamma P_\sigma) \qquad (\lambda \neq 0). \tag{3}$$

4.2 One block is one rectangular product

Since Γ^{-1} and ΓP_σ are matrices with entries in the coefficient field, (3) transforms all three noncommutative forms of a block word at once: with $X' = (\Gamma^{-1})^{\mathsf{T}} X$ and $Y' = \Gamma P_\sigma Y$,

$$f^{xy}_{A+\sigma} = \lambda\, f^{xy}_{A'}(X', Y'), \quad f^{yx}_{A+\sigma} = \lambda\, f^{yx}_{A'}(X', Y'), \quad f_{A+\sigma} \star z = \lambda\, (f_{A'} \star z)(X', Y').$$

Each gate factor $c_1 + c_2 P$ has two nonzero entries per row, so X' and Y' cost $O(m2^n)$ gates. Since A' is supported on the first m qubits, $P_{A'} = P_{\alpha,\beta}$ with α, β supported there; splitting $x = (p, s) \in \mathbb{F}_2^m \times \mathbb{F}_2^{n-m}$,

$$f^{xy}_{\alpha,\beta}(X', Y') = \sum_p (-1)^{\beta \cdot p} M[p, p+\alpha], \qquad M[p, q] = \sum_{s \in \mathbb{F}_2^{n-m}} X'(p, s)\, Y'(q, s).$$

So the xy-forms of all words of the block are Walsh coefficients along shifted diagonals of one product M of a $2^m \times 2^{n-m}$ and a $2^{n-m} \times 2^m$ matrix; the yx-forms come from the swapped product M'. Grouping the block's words by their canonical label (α, β), the middle-insertion terms combine into one trilinear form $\sum_{p,s,q} X'(p, s)\, Q[p, q]\, Y'(q, s)$ with $Q[p, q] = \sum_b (-1)^{p \cdot b} g(p + q, b)$ for suitable values g.

Definition 4.2. A *block oracle* with cost M for $2^m \times 2^r$ by $2^r \times 2^m$ products is a family of circuits that, from arbitrary available values $X'(p, s)$, $Y'(q, s)$ and $Q[p, q]$ in the free algebra ($p, q \in \mathbb{F}_2^m$, $s \in \mathbb{F}_2^r$), compute all entries $\sum_s X'(p, s) Y'(q, s)$, and the single value $\sum_{p,s,q} X'(p, s) Q[p, q] Y'(q, s)$, each with at most M further gates.

Proposition 4.3. *Given a block oracle with cost M, the sum $\sum_\nu c_\nu f_\nu^{[2]}$ over all words of a block, with arbitrary coefficients, takes at most $O(m2^n) + 3M + 15(m+1)4^m + 2$ gates, independently of the number of words.*

Proof. Products and swapped products: $2M$. Walsh–Hadamard transforms of all 2^m shifted diagonals: $O(m4^m)$ via the Kronecker structure. Squares and the regrouped coefficients: $O(4^m)$. Middle insertion: M, after $O(m4^m)$ to form Q. □

Block oracles come from two sources.

(i) *Programs (over $\mathbb{C}$).* A correct straight-line program of cost s for the $2^m \times 2^{n-m} \times 2^m$ product gives a block oracle with cost $960s$. Each register is split into its constant, x-linear, y-linear and *ordered* xy-bilinear parts (the product rule $\partial^2(pq) = p(0)\partial^2 q + q(0)\partial^2 p + \partial_x p\, \partial_y q + \partial_x q\, \partial_y p$ keeps the x factor on the left), which yields all ordered products with $91s$ gates; the middle insertion is obtained by reverse differentiation of the program, in the spirit of Baur–Strassen [5] and Hrubeš's Lemma 20 [10]. The polynomial semantics uses that $\mathbb{C}$ is infinite.

(ii) *Rank decompositions (any field).* A decomposition $\langle a, b, a \rangle = \sum_{r \le R} u_r \otimes v_r \otimes w_r$ gives the classical recursive bilinear algorithm, which is valid in any noncommutative algebra because scalars are central and x-factors stay on the left; the trilinear identity $\sum_{i,j,l} X_{ij} Q_{il} Y_{jl} = \sum_r \big(\sum u_r X\big)\big(\sum w_r Q\big)\big(\sum v_r Y\big)$ recurses the same way. With t levels this costs at most $(7(2ab + a^2) + 6)(t+1) \max(R, ab, a^2)^t$.

5 Covering the corner classes

Fix $\frac{1}{2} < \mu < 1$ and $m = \lceil \mu n \rceil$. The words of class (a, b) are exactly those with Majorana vector $A + \sigma_{a,b}$, where $\sigma_{a,b}$ complements the symmetric part if $a > n - a$ and the skew part if $b > n - b$, and A has weights $(a_0, b_0) = (\min(a, n-a), \min(b, n-b))$. A block with u_s symmetric and u_a skew Majoranas ($u_s + u_a = 2m$) contains $\binom{u_s}{a_0}\binom{u_a}{b_0}$ of them.

Lemma 5.1 (greedy cover)**.** *For $a_0 \le u_s \le n$ and $b_0 \le u_a \le n$ there is a family of at most*

$$1 + 2n \, \frac{\binom{n}{a_0}\binom{n}{b_0}}{\binom{u_s}{a_0}\binom{u_a}{b_0}}$$

blocks (U_s, U_a) with $|U_s| = u_s$, $|U_a| = u_a$ such that every pair (S, T) with $|S| = a_0$, $|T| = b_0$ has $S \subseteq U_s$, $T \subseteq U_a$ for some block of the family.

Proof. Every pair lies in $D = \binom{n-a_0}{u_s-a_0}\binom{n-b_0}{u_a-b_0}$ of the $\binom{n}{u_s}\binom{n}{u_a}$ blocks, and $\binom{n}{u}\binom{u}{a_0} = \binom{n}{a_0}\binom{n-a_0}{u-a_0}$. The greedy algorithm covers at least a fraction $D/\#\text{blocks}$ of the remaining pairs at each step, so $1 + (\#\text{blocks}/D) \ln \#\text{pairs}$ blocks suffice, and $\ln \#\text{pairs} \le 2n$. □

Lemma 5.2 (layout)**.** *Let $g_\mu = 2\mathrm{H}(\frac{1}{4}) - 2\mu\mathrm{H}(\frac{1}{4\mu})$. For every $\delta > 0$ and all large n, every corner class $(a, b) \in \mathcal{C}_n$ admits $u_s + u_a = 2m$, $u_s, u_a \le n$, $a_0 \le u_s$, $b_0 \le u_a$, with*

$$\binom{n}{a}\binom{n}{b} \le 2^{n(g_\mu+\delta)} \binom{u_s}{a_0}\binom{u_a}{b_0}.$$

Proof sketch. Write $L = 2\mu$, $\alpha = a_0/n \le \beta = b_0/n$, $t = \alpha + \beta \le \frac{1}{2}$. If $\beta/t \le 1/L$, take $u_s/n = L\alpha/t$, $u_a/n = L\beta/t$; concavity of entropy bounds the ratio by $2^{n(2\mathrm{H}(t/2) - L\mathrm{H}(t/L))}$, and $2\mathrm{H}(t/2) - L\mathrm{H}(t/L)$ is nondecreasing in t, so it is at most g_μ. If $\beta/t > 1/L$, take $u_a = n$, $u_s = (L-1)n$; the exponent $\mathrm{H}(\alpha) - (L-1)\mathrm{H}(\alpha/(L-1))$ is increasing in α and agrees with the first case on the boundary. The formal proof works with exact binomials: a Vandermonde-type identity $\sum_u \binom{u}{a_0}\binom{M-u}{b_0} = \binom{M+1}{a_0+b_0+1}$ chooses the integer layout, the bounds $\binom{N}{K}K^K(N-K)^{N-K} \le N^N \le (N+1)\binom{N}{K}K^K(N-K)^{N-K}$ convert to entropy with only a polynomial loss, and convexity of $x \log x$ reduces the exponent to the extreme case $a_0 + b_0 = \lfloor n/2 \rfloor$; the threshold is $n_0 = \lceil 54/(\delta \ln 2)^3 \rceil + 1$. □

6 Assembling the circuit

Proof of Theorem 1.4. Let $N = 2^n$. By Theorem 3.2 and Lemma 2.2, $\mathrm{ID} = \sum_\nu w_\nu f_\nu^{[2]}$ with w supported on corner classes. For each corner class choose the layout of Lemma 5.2 and the cover of Lemma 5.1; assign each word to one covering block. By Proposition 4.3 each block costs $O(m2^n) + 3M + O(m4^m)$, with $M \le C_\varepsilon(2^m)^{w+\varepsilon}$ by hypothesis. There are at most $(n+1)^2$ classes and, per class, at most $1 + 2n2^{n(g_\mu+\delta)}$ blocks. Since $\mu > \frac{1}{2}$ and $w \ge 2$, $2^n \le 4^m \le (2^m)^w$, so the total is

$$2^{n(g_\mu+\delta)+\mu n(w+\varepsilon)+O(\log n)} = N^{C(\mu,w)+o(1)}.$$

Arbitrary N follows by padding to the next power of two. □

7 Matrix multiplication inputs

Both routes supply the block oracles of Theorem 1.4 from one fixed rank decomposition, through source (ii) of Section 4, and both pass to the base field through the regular representation.

Lemma 7.1 (one decomposition suffices)**.** *Let $\frac{1}{2} < \mu < 1$ and $k = \frac{1-\mu}{\mu}$. Let E be a field with $2 \neq 0$ that contains a square root of -1. Suppose that over E the tensor $\langle a, b, a\rangle$, with $a \geq 2$ and $b \geq a^k$, has rank at most ρ, and that $\max(\rho, ab, a^2) \leq a^\tau$. Then $\mathrm{size}_E^{\mathrm{nc}}(\mathrm{ID}_N) \leq N^{C(\mu,\tau)+o(1)}$.*

Proof. Let $m = \lceil \mu n \rceil$ and let t be least with $a^t \geq 2^m$. Since $n - m \leq mk$, also $b^t \geq a^{kt} \geq 2^{n-m}$, so the block products of size $2^m \times 2^{n-m} \times 2^m$ embed, by padding with zeros, in $\langle a^t, b^t, a^t\rangle$. Source (ii) gives block oracles with cost at most $(7(2ab + a^2) + 6)(t + 1)\, a^{t\tau}$, and $a^{t\tau} \leq a^\tau (2^m)^\tau$. Since $t = O(m)$, this is $O((2^m)^{\tau+\varepsilon})$ for every $\varepsilon > 0$, and Theorem 1.4 applies. □

Lemma 7.2 (descent)**.** *If $E/\mathbb{F}$ is a finite extension of degree d, then $\mathrm{size}_{\mathbb{F}}^{\mathrm{nc}}(\mathrm{ID}_N) \leq d^2(2d + 1)\, \mathrm{size}_E^{\mathrm{nc}}(\mathrm{ID}_N)$.*

Proof. E embeds in $d \times d$ matrices over $\mathbb{F}$ by its regular representation, so a circuit over E becomes a circuit over $\mathbb{F}$ computing a $d \times d$ matrix of noncommutative polynomials, each gate simulated entrywise with at most $d^2(2d + 1)$ gates. A diagonal entry of the image of ID is ID itself. □

7.1 Uznański's curve over the integers

Let $f(k) = 2$ for $0 \leq k \leq \frac{1}{2}$ and $f(k) = 1 + k + \frac{1}{4k}$ for $k > \frac{1}{2}$. Uznański [33] proved $\omega(1, k, 1) \leq f(k)$; in particular $\alpha \geq \frac{1}{2}$ and $\omega \leq \frac{9}{4}$, the bound OpenAI had proved over $\mathbb{C}$ shortly before [23]. Zhou and Wang [34] extended the bound to every associative ring, with integer bilinear algorithms, and formalized the extension in Lean. We use one theorem of their formalization.

Theorem 7.3 (Uznański [33]; Zhou and Wang [34])**.** *For every $k \geq 0$ and $\varepsilon > 0$ there are integers $a \geq 2$, $b \geq a^k$ and $\rho \geq \max(ab, a^2)$, and integer vectors u_r, v_r, w_r $(r \leq \rho)$, such that $\langle a, b, a\rangle = \sum_{r\leq\rho} u_r \otimes v_r \otimes w_r$ over $\mathbb{Z}$ and $\log \rho / \log a < f(k) + \varepsilon$.*

In Lean this is `Matmul.RectangularRankBridge.exists_scheme_certificate`. Its proof bounds the asymptotic rank of integer matrix tensors through spectral points in the sense of Strassen [31], following Uznański's shared-leg entropy argument, and extracts a single decomposition with the given slack.

Proof of Theorem 1.1. Take $\mu = \frac{18}{29}$. Then $k = \frac{1-\mu}{\mu} = \frac{11}{18}$ and $f(k) = 1 + \frac{11}{18} + \frac{9}{22} = \frac{200}{99}$. Let $\varepsilon_0 = 10^{-5}$ and fix the decomposition of Theorem 7.3 for this k and $\varepsilon = \varepsilon_0$.

Let $\mathbb{F}$ have characteristic other than two, and let E be $\mathbb{F}$ with a root of $X^2 + 1$ adjoined (or $E = \mathbb{F}$ if $\mathbb{F}$ already has one), so $[E : \mathbb{F}] \leq 2$. The integer decomposition maps into E, and $\max(\rho, ab, a^2) = \rho \leq a^\tau$ with $\tau = \frac{200}{99} + \varepsilon_0$. Lemma 7.1 gives $\mathrm{size}_E^{\mathrm{nc}}(\mathrm{ID}_N) \leq N^{C(\mu,\tau)+o(1)}$, where, since C is affine in w,

$$C(\mu, \tau) = C\big(\tfrac{18}{29}, \tfrac{200}{99}\big) + \tfrac{18}{29}\varepsilon_0 = 1.66916842\ldots + 0.0000062\ldots < 1.6692.$$

The value $C(\frac{18}{29}, \frac{200}{99}) + 10^{-5} < 1.6692$ is certified with rational enclosures of $\log 2$, $\log \frac{3}{2}$, $\log \frac{36}{29}$ and $\log \frac{72}{43}$. Lemma 7.2 transfers the bound to $\mathbb{F}$, and Hrubeš's reductions give $S_{4,N}$ and $\mathrm{perm}_{4,N}$. □

The choice $\mu = \frac{18}{29}$ is a rational point near the optimum: over μ, the exponent with $w = f(k)$ is minimized near $\mu \approx 0.6207$, at $1.669168\ldots$.

7.2 OpenAI's bounds

Theorem 7.4 (OpenAI [24, 25]). *Over $\mathbb{C}$: $\alpha > 0.465$, so $\omega(1, \frac{93}{200}, 1) = 2$; and $\omega(1, 0.709, 1) < \frac{523}{250} = 2.092$.*

These are formalized in OpenAI's Lean library, which we import unchanged. The first is used through the strict supremum: some aspect $k > 0.465$ has $\omega(1, k, 1) = 2$, and admissibility is monotone in the aspect.

Lemma 7.5 (programs give ranks). *A correct straight-line program of cost s for $a \times b$ by $b \times c$ matrix multiplication over $\mathbb{C}$ yields a rank decomposition of the matrix-multiplication tensor $\langle a, b, c\rangle$ of size $2s$.*

Proof. This is the classical comparison of multiplicative and bilinear complexity (Strassen [32]; see [6, Ch. 14]): by the product rule above, the bilinear part of every register is a linear combination of the $2 \cdot \#\{\times \text{ gates}\}$ rank-one forms $\partial_x p \otimes \partial_y q$, $\partial_x q \otimes \partial_y p$, and correctness identifies the bilinear parts of the outputs with the tensor. □

Lemma 7.6 (rational interpolation). *If $\tau_i \geq 2$ is admissible at aspect $k_i \in [0, 1]$ $(i = 1, 2)$ over $\mathbb{C}$, then for $p, q \in \mathbb{N}$ with $p + q > 0$ the exponent $\frac{p\tau_1 + q\tau_2}{p+q}$ is admissible at aspect $\frac{pk_1 + qk_2}{p+q}$.*

This is a special case of the classical convexity of $\omega(1, k, 1)$ (Lotti and Romani [18]).

Proof. Fix $\varepsilon > 0$. For large t take programs for sizes $n_1 = 2^{tp}$ and $n_2 = 2^{tq}$ at the two endpoints, convert them to rank decompositions (Lemma 7.5), and take their tensor product: a decomposition of $\langle a, b, a\rangle$ with $a = 2^{t(p+q)}$, $b \geq a^k$, and rank at most $4C_1C_2\, a^{\tau+\varepsilon} \leq a^{\tau+2\varepsilon}$ once t is large. The block recursion ($\langle a, b, a\rangle\otimes$ a program for any size) turns this single decomposition into an admissible exponent $\tau + 2\varepsilon$. □

With $p = 829933$ and $q = 716295$, the endpoints $(\frac{93}{200}, 2)$ and $(\frac{709}{1000}, \frac{523}{250})$ interpolate to aspect $k^* = \frac{3663}{6337} = \frac{1-\mu}{\mu}$ for $\mu = \frac{6337}{10000}$, with exponent $w^* = \frac{157917757}{77311400} = 2.0426192\ldots$. Then

$$C(\mu, w^*) = 1.69056770515\ldots < 1.690568 - 10^{-7},$$

certified with rational enclosures of $\log 2$, $\log \frac{3}{2}$, $\log \frac{10000}{6337}$ and $\log \frac{7674}{6337}$. This proves Theorem 1.2(a) over $\mathbb{C}$; every field of characteristic zero is covered by (b), whose set S contains only primes.

Remark 7.7. The same formula with only the rectangular bounds of Alman et al. [1] gives $C < 1.7118$, and OpenAI's thin endpoint alone ($\mu = \frac{200}{293}$, $w = 2$) gives $1.6938884\ldots$; the latter is also formalized.

7.3 Other characteristics for OpenAI's route

Proof of Theorem 1.2(b). Let $\varepsilon_0 = 10^{-8}$. The proof of Lemma 7.6 produces one rank decomposition over $\mathbb{C}$ of a fixed tensor $\langle a, b, a\rangle$, with $a^{k^*} \leq b$ and $\max(R, ab, a^2) \leq a^{w^*+2\varepsilon_0}$. Its existence is an existential first-order sentence in the language of rings with integer coefficients, true in $\mathbb{C}$. By completeness of the theory of algebraically closed fields of characteristic zero and the Lefschetz principle, it holds in every algebraically closed field of characteristic 0 or $p \notin S$, for a finite set S of primes.

Let $\mathbb{F}$ have characteristic 0 or $p \notin S \cup \{2\}$. The decomposition exists over the algebraic closure; its finitely many coefficients and a square root of -1 generate a finite extension $E/\mathbb{F}$. Lemma 7.1 gives $\mathrm{size}_E^{\mathrm{nc}}(\mathrm{ID}_N) \leq N^{C(\mu, w^*+2\varepsilon_0)+o(1)}$, and $C(\mu, w^* + 2\varepsilon_0) = C(\mu, w^*) + 2\mu\varepsilon_0 < c_0$. Lemma 7.2 transfers the bound to $\mathbb{F}$, and Hrubeš's reductions give $S_{4,N}$ and $\mathrm{perm}_{4,N}$. □

The argument is not specific to the numerical choice.

Theorem 7.8 (parametric field transfer)**.** *Let $\frac{1}{2} < \mu < 1$ and let w be an admissible rectangular exponent over $\mathbb{C}$ at aspect $\frac{1-\mu}{\mu}$.*

(a) If $c \geq 1$ and $C(\mu, w) < c$, there is a finite set S of primes such that the three circuit bounds hold at exponent c over every field of characteristic zero or characteristic outside $S \cup \{2\}$. The set S does not depend on the field or on the slack ε in the circuit bound.

(b) If $C(\mu, w) \geq 1$, the three bounds hold at the exact exponent $C(\mu, w)$ over every field of characteristic zero, whether or not it contains a square root of -1.

Proof. For (a), put $\eta = (c - C(\mu, w))/(4\mu) > 0$. The proof of Lemma 7.6, with one endpoint, gives one rank decomposition with exponent at most $w + 2\eta$; transfer it and descend as above. The resulting exponent is $C(\mu, w + 2\eta) = C(\mu, w) + 2\mu\eta = \frac{1}{2}(C(\mu, w) + c) < c$. For (b), apply (a) with $c = C(\mu, w) + \delta$ for every $\delta > 0$; a field of characteristic zero avoids every exceptional set. Given slack ε, use $\delta = \varepsilon/2$ and slack $\varepsilon/2$. □

Part (b) does not give one finite exceptional set in positive characteristic at the exact exponent: the sets produced as $\delta \to 0$ need not have a finite union.

No transfer of $\alpha > 0.465$ itself to positive characteristic is known; OpenAI states explicitly that none is asserted [24, §14], because α requires decompositions of every accuracy. This argument needs only one, because the exponent has slack. This is also why S is not computed. Theorem 1.1 makes the question moot for the circuit bound, since its decompositions have integer coefficients from the start.

8 Formal verification

8.1 What is checked

The development uses Lean 4.34.1 and Mathlib (commit `d13f23b`). It consists of

- OpenAI's matrix-multiplication library [25] (509 files from `openai/math`, commit `7ef5c0c`), unchanged, with their SHA-256 hashes recorded;
- the `Matmul` library of Zhou and Wang [34] (64 files, commit `285f513`), with its license and notice files. Two of its declarations relied on Lean's automatic implicit arguments, which our build disables; we added the missing binder `{t : Nat}` to each, checked that the resulting types and bodies are identical to the upstream elaboration, and marked the file as changed. Nothing else was changed;
- harfe's formalization of Brouwer's fixed-point theorem [8], a dependency of both libraries, adapted to our Lean and Mathlib versions;
- the `PauliCubes` library (about 4,700 lines; first written for Lean 4.33.1 and ported with the edits recorded in the repository), which proves Theorem 1.3, Hrubeš's Lemma 22 in the form of Lemma 2.2, and the reductions for $S_{4,N}$ and $\mathrm{perm}_{4,N}$;
- the `IDMM` library (51 files, about 8,000 lines) for this paper.

The top-level statements are:

- `IDMM.id_bound_every_char` (Theorem 1.1): for every field `F : Type u` with `ringChar` $F \neq 2$, `IDBoundF`, `S4BoundF` and `Perm4BoundF` hold over `F` at $4173/2500 = 1.6692$;
- `IDMM.main_target : MainTarget`, i.e. `IDBound`, `S4Bound` and `Perm4Bound` at $211321/125000$ over $\mathbb{C}$;
- `IDMM.main_target_almost_every_char`: there is a finite `S : Finset Nat` such that for every field `F : Type u` with `ringChar` $F \neq 2$ and `ringChar` $F = 0 \vee$ `ringChar` $F \notin S$, the same three bounds hold over `F`;
- `IDMM.main_target_char_zero`: the same bounds over every field of characteristic zero;
- `PauliCubes.id_circuit`, `PauliCubes.s4_circuit` and `PauliCubes.perm4_circuit` (Theorem 1.3): over every field with $2 \neq 0$ and for every $\varepsilon > 0$, circuits of size $O(N^{1.727899+\varepsilon})$; `IDMM.cubes_bound` restates them as `IDBoundF`, `S4BoundF` and `Perm4BoundF`;
- `IDMM.construction_almost_every_char` and `IDMM.construction_char_zero`: Theorem 7.8(a) and (b);
- `IDMM.thin_main`: the bound $16939/10000$ over $\mathbb{C}$ from the thin endpoint alone;
- `IDMM.construction_target : ConditionalTarget` (Theorem 1.4 over $\mathbb{C}$, for every $\mu \in (\frac{1}{2}, 1)$ and every admissible w) and `IDMM.construction_gen` (any field with $2 \neq 0$ and $\sqrt{-1}$, given block oracles);
- `IDMM.strict_id_bound_every_char`, `IDMM.strict_main_target`, `IDMM.strict_main_target_char_zero`, `IDMM.strict_cubes_bound` and `IDMM.strict_main_target_almost_every_char`: Theorems 1.1, 1.2 and 1.3 in the strict circuit model described in Section 8.2;
- `IDMM.two_mul_le_of_ncSizeLE_ID`, `IDMM.not_idBoundF_of_lt_one`, `IDMM.id_bound_every_char_rat` and `IDMM.id_bound_every_char_zmod3`: the non-vacuity checks described there.

Every one of them depends only on Lean's axioms `propext`, `Classical.choice` and `Quot.sound`; no `sorry`, `native_decide` or added axiom occurs anywhere.

The repository `https://github.com/research-notes-100/identity-polynomial-circuits` contains exactly the sources verified. On October 29, 2026, from 10:16 to 10:48 p.m. Eastern, `scripts/validate.py` ran on them, in order: the full `lake build`; the audit modules `IDMM.FinalAudit` and `IDMM.FinalAuditEveryChar`, which print the signature and axiom list of every theorem above (`scripts/audit.py` fails if one is missing); and `leanchecker --fresh` on both modules, which replays each module and everything it imports through the kernel in a fresh environment. Every stage exited with status 0. The record in `validation/release` gives the SHA-256 hash of every Lean source, pin file and log, and `scripts/audit.py` checks the repository against it and checks OpenAI's and Zhou and Wang's files against their upstream hashes. The `theorem-b` directory, a separate Lake project on Lean 4.33.1 and Anthropic's formalization, was checked the same day from 4:13 to 4:52 p.m.: a full build, a check that each of its 312 constants uses only the three axioms above, and kernel replays of all seven of its modules, in place and with `--fresh` (`validation/theorem-b`).

Zhou and Wang's library and OpenAI's library declare some of the same names (integer and complex versions of shared modules), so they cannot be imported into one file. The proof of Theorem 1.1 therefore uses Zhou and Wang's library together with the OpenAI modules it needs, but not OpenAI's top-level matrix-multiplication theorems; the proofs of Theorem 1.2 use OpenAI's library and not Zhou and Wang's. Each closure is built and checked on its own. The import closure of `IDMM.id_bound_every_char` contains 52 modules of Zhou and Wang's library, 55 of OpenAI's, 5 of harfe's, 22 of `PauliCubes` and 34 of `IDMM`, besides Mathlib.

8.2 The trusted definitions

A formal theorem is only as good as its statement. The statements rest on the following definitions, which we compared with Hrubeš's [10].

- *Circuits.* `NCSizeLE f s` holds if some list of at most s elements of the free algebra $\mathbb{F}\langle V\rangle$, each a variable, a constant of $\mathbb{F}$, or the sum or product (in either order) of two earlier elements, contains f. A list of node values in topological order is the same as a circuit in the sense of Definition 2.1, counting inputs as nodes; the only difference is that a gate may use the same earlier node twice. Read strictly, Definition 2.1 requires the two inputs of a gate to be different nodes. `StrictNCSizeLE` formalizes that reading by node positions, and `IDMM.ncSizeLE_strict` turns a circuit with s nodes into a strict one with at most $2s+1$ nodes: one constant node 1, and after every node g a copy $1 \cdot g$. Conversely every strict circuit is a circuit (`IDMM.StrictNCSizeLE.ncSizeLE`). Theorems 1.1, 1.2 and 1.3 are proved in the strict model as well.

- *Non-vacuity.* Every circuit computing ID_N has at least $2N$ nodes, one for each variable (`IDMM.two_mul_le_of_ncSizeLE_ID`). Hence `IDBoundF F c` is false for every $c < 1$, in both models (`IDMM.not_idBoundF_of_lt_one`, `IDMM.not_strictIDBoundF_of_lt_one`): no bound statement holds with an exponent below 1. These lower bounds do not show that our exponents are optimal. Theorem 1.1 is also instantiated over $\mathbb{Q}$ and over $\mathbb{F}_3$, so its hypothesis is satisfiable.

- *Polynomials.* `ID N`, `S4 N` and `perm4 N` are exactly the sums displayed in Section 2.

- *Fields.* The theorems over general fields quantify over `F : Type u` for an arbitrary universe `u`.

- *Bounds.* `IDBound c` (resp. `IDBoundF F c`) states: for every $\varepsilon > 0$ there is $C > 0$ with `NCSizeLE (ID N)` $\lfloor CN^{c+\varepsilon} \rfloor$ for all $N \geq 1$.

- *Matrix multiplication.* The statement of Theorem 1.1 mentions no matrix-multiplication notion; the input enters only inside the proof, as an integer decomposition checked against Zhou and Wang's `matrixTensor`, which `IDMM.rankAtMost_of_scheme` maps into any commutative ring. The notions of Section 2.4, used for Theorem 1.2, are OpenAI's definitions, imported unchanged.

The statement file of the conditional milestone (`IDMM/Statements.lean`), the circuit definitions and OpenAI's files are unchanged from the versions in which they were first written.

9 Discussion

(1) *Every odd characteristic.* Theorem 1.1 is uniform across characteristics, with no exceptional odd primes: one integer decomposition serves every field. (It is an existence proof; neither we nor Zhou and Wang extract an executable algorithm.) The uncomputed set S of Theorem 1.2(b) remains only in that route. Lynn [19] studied the analogous question for sum-of-squares formulas over algebraically closed fields.

(2) *Better matrix multiplication feeds in directly.* The exponent is $C(\mu, w)$ with w any rectangular exponent at aspect $\frac{1-\mu}{\mu}$, and the formal Lemma 7.1 accepts any rank decomposition. An improvement of $\omega(1, k, 1)$ near $k \approx 0.61$, given by one explicit decomposition, would improve Theorem 1.1 with no new circuit argument.

(3) *Characteristic two* is excluded by Lemma 2.2, which divides by 2.

(4) *The floor of the cost formula* is $2\mathrm{H}(\frac{1}{4}) = 1.6226$, approached only if $\omega = 2$. It bounds this accounting, not circuits: the count is of all words in corner classes, so an identity with fewer nonzero weights, or evaluation shared across blocks, could go below it, as could a different route from sums of squares to circuits.

(5) *Lower bounds.* Nothing here bears on lower bounds; the program of [11] needs only $\sigma(n) \geq n^{1+\varepsilon}$.

Acknowledgments

This work rests on the work of many people, and it is a pleasure to thank them.

Pavel Hrubeš, Avi Wigderson and Amir Yehudayoff created the bridge between sums of squares and noncommutative circuits on which everything here stands, and Hrubeš's 2024 paper supplied both the first subquadratic bounds and the exact form of the reduction (his Lemma 22) that we use. Behind them stand Hurwitz and Radon, and Daniel Shapiro's monograph, which keeps the classical theory accessible. Noam Nisan introduced the noncommutative circuit model. The algebraic tools are Pascual Jordan and Eugene Wigner's 1928 transformation, Mykhailo Krawtchouk's polynomials, and the Baur–Strassen theorem.

The main theorem rests on Przemysław Uznański's rectangular bound [33], which its author reports was produced with OpenAI's Codex, and on Yang Zhou and Mingsheng Wang's extension of it to every ring, with its Lean formalization, released openly [34]. Their integer decompositions are what remove every restriction on the characteristic beyond Hrubeš's. Their development builds on OpenAI's library, on Jeroen Zuiddam's work on asymptotic spectra [35] and his Lean development `omega-nine-quarters-all-fields` [36], and on harfe's formalization of Brouwer's fixed-point theorem [8]; the spectral method goes back to Strassen [31].

The rectangular matrix-multiplication bounds of the second route, and the 9/4 bound that started this line, are due to OpenAI [24, 23], who also formalized them in Lean and released the formalization openly [25]; we import 509 of their files unchanged, and without that library the second route would rest on an unverified input. Their work builds on a long line, including Volker Strassen [30, 32], Arnold Schönhage [27], Don Coppersmith and Shmuel Winograd [7], Grazia Lotti and Francesco Romani [18], François Le Gall and Florent Urrutia [17], and Josh Alman, Ran Duan, Virginia Vassilevska Williams, Yinzhan Xu, Zixuan Xu and Renfei Zhou [1]; Amir Shpilka's exposition [29] helped us read it. This project grew out of studying the October 2026

work of OpenAI and of Anthropic on the Exponential Time Hypothesis. Anthropic's paper [3] and its Lean-verified `formal-math` development [4] led us to the improved composition bound mentioned in the introduction and from there to this problem; its single-bit analysis of the crossing tensor anticipates our first Pauli decomposition; and it set the standard of machine-checked proof that we have tried to meet.

The formal proofs depend on Lean 4, created by Leonardo de Moura and Sebastian Ullrich and developed by the Lean FRO and its contributors, and on Mathlib, whose community built, among much else, the model theory of algebraically closed fields and the Lefschetz principle used in Section 7.3. We thank A. Kapila and Vaibhav Tulsyan, whose independent same-day notes [14, 37] on sums of squares we became aware of during this work.

Use of AI systems. The research and the formalization were carried out by the author working with AI systems: OpenAI's Codex (in ChatGPT) for research cycles, including the construction of Sections 3–6, and for a review of the formalization that contributed Theorem 7.8 and its Lean proof; and Anthropic's Claude for review, the Lean formalization and the drafting of this paper. All theorems and lemmas are machine-checked as described in Section 8; the author is responsible for the contents.

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